

S_3 -equivariant jet filtration of the three-body Poisson algebra: tier decomposition, integer scaling exponents, and syzygy structure

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Abstract

In a companion paper we showed that the Poisson algebra $\mathcal{A}$ generated by the three pairwise Hamiltonians of the planar three-body problem has dimension sequence $d(0) = 3$, $d(1) = 6$, $d(2) = 17$, $d(3) = 116$, with 156 candidate generators at level 3 of which 116 are linearly independent. Here we reveal the internal organisation of this algebra. The singular value (SV) spectrum at each configuration exhibits a four-tier staircase

$$52 + 44 + 16 + 4 = 116$$

with inter-tier gaps spanning factors of 10^2 – 10^4 (two to four orders of magnitude). We prove that the tier sizes are determined by the S_3 isotypic decomposition of the algebra: the 156 generators carry exactly 24 copies of the trivial representation A , 28 copies of the sign representation A' , and 52 copies of the standard representation E , and the number $n_E = 52$ equals the first tier size exactly. The Clebsch–Gordan rules of S_3 predict the multiplicities at every bracket level.

We identify a jet filtration underlying the tier structure: fitting singular values as $\sigma \sim \varepsilon^\alpha$ across five sampling radii yields integer-quantized scaling exponents $\alpha = 0, 1, 2, 3$ for tiers 1–4 respectively, with the noise floor flat at machine epsilon. The tiers thus correspond to successive orders of the Taylor expansion of generator functions in configuration space. The gap at the rank boundary grows as $\sim \varepsilon^2$, from $5.5\times$ at $\varepsilon = 10^{-4}$ to $67,472\times$ at $\varepsilon = 2 \times 10^{-3}$.

The 40 null directions in the SV spectrum are not Dirac constraints but syzygies: all 156 generators are individually nonzero as phase-space functions, and the null space decomposes into 8 universal identities (machine-epsilon everywhere) and 32 soft syzygies, 8 of which break at the collinear submanifold (raising the rank from 116 to 124), with an atlas-wide maximum of 125 (9 awake) at a single grid cell. The tier decomposition and the syzygy structure are orthogonal. All results are potential-independent: the $1/r$ and $1/r^2$ systems share the same tier sizes, scaling exponents, and null counts.

1 Introduction

In a companion paper [1] we studied the Poisson algebra $\mathcal{A}$ generated by the three pairwise interaction Hamiltonians H_{12} , H_{13} , H_{23} of the planar three-body problem. The cumulative dimension through bracket level n ,

$$d(0) = 3, \quad d(1) = 6, \quad d(2) = 17, \quad d(3) = 116,$$

grows super-exponentially, consistent with conjectured infinite Gelfand–Kirillov dimension. The sequence is invariant under changes of mass ratios and identical for the Newtonian ($1/r$) and Calogero–Moser ($1/r^2$) potentials, establishing it as an algebraic invariant of the exceptional-exponent classification.

That analysis characterised the algebra by its *dimension*. The present paper asks a finer question: *how is the 116-dimensional space internally organised?* We find three interlocking structures:

1. A **four-tier decomposition** $52 + 44 + 16 + 4 = 116$, visible as shelves in the singular value spectrum separated by gaps of 10^2 – 10^4 (Section 3).
2. An S_3 **isotypic decomposition** that explains the tier sizes exactly via Clebsch–Gordan rules (Section 4).
3. A **jet filtration** in which the tiers correspond to orders 0, 1, 2, 3 of the local Taylor expansion, with integer-quantized scaling exponents (Section 5).

We also resolve the status of the 40 null directions in the SV spectrum: they are syzygies among iterated Poisson brackets, not Dirac constraints (Section 6). All structures are universal across singular potentials and configurations.

2 Setup and notation

We follow the conventions of [1]. Three point masses m_1, m_2, m_3 in the plane have phase space $\mathbb{R}^{12}$ with canonical coordinates $(x_i, y_i, p_{x_i}, p_{y_i})$, $i = 1, 2, 3$. The pairwise Hamiltonians are

$$H_{ij} = \frac{p_{x_i}^2 + p_{y_i}^2}{2m_i} + \frac{p_{x_j}^2 + p_{y_j}^2}{2m_j} + V(r_{ij}),$$

where $V(r) = -1/r$ (Newtonian) or $V(r) = -1/r^2$ (Calogero–Moser). Using the polynomial representation $u_{ij} = 1/r_{ij}$, all Poisson brackets are computed in exact rational arithmetic (levels 0–3) or numerically (level 4).

2.1 Generator census

The algebra $\mathcal{A}$ is generated by $\{H_{12}, H_{13}, H_{23}\}$ under iterated Poisson brackets. At each level n , the new generators are all brackets $\{f, g\}$ with f from the frontier at level $n - 1$ and g from the accumulated set through level $n - 1$, modulo antisymmetry. The candidate counts are:

Level	Construction	Candidates	New generators
0	H_{12}, H_{13}, H_{23}	3	3
1	$\{\mathcal{A}_0, \mathcal{A}_0\}$	3	3
2	$\{\mathcal{A}_1, \mathcal{A}_0\} + \Lambda^2 \mathcal{A}_1$	12	11
3	$\{\mathcal{A}_2, \mathcal{A}_0\} + \{\mathcal{A}_2, \mathcal{A}_1\} + \Lambda^2 \mathcal{A}_2$	138	99
Total		156	116

Of the 156 candidate generators, 116 are linearly independent at generic phase-space points and 40 satisfy algebraic relations. The number 156 is the total candidate count (all non-antisymmetric bracket pairs); it is not the number of nonzero expressions.

2.2 Numerical evaluation

To probe the algebra at a specific configuration q in shape space, we sample N phase-space points uniformly in an ε -ball around q and evaluate all 156 generators to form the evaluation matrix $M \in \mathbb{R}^{N \times 156}$. After column normalisation, the singular value decomposition $M = U\Sigma V^T$ yields the sorted singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{156}$. The effective rank is determined by the largest gap in the σ -spectrum.

We parametrise the reduced configuration space (shape sphere for equal masses) by the mass ratio $\mu = m_3/m_1$ and the angle ϕ measuring departure from collinear alignment. A 100×100 grid over $(\mu, \phi) \in [0.2, 3.0] \times [3^\circ, 177^\circ]$ provides 10,000 configurations, each evaluated at five sampling radii $\varepsilon \in \{10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$.

3 The tier decomposition

3.1 Observation: a universal staircase

At every non-degenerate configuration, the normalised SV spectrum σ_k/σ_1 displays a characteristic staircase pattern with four shelves separated by abrupt drops.

Proposition 1 (Tier structure). *For both the $1/r$ and $1/r^2$ potentials, at every configuration with $\phi > 10^\circ$, the 156-component SV spectrum evaluated at $\varepsilon = 10^{-4}$ exhibits four tiers:*

<i>Tier</i>	<i>SV indices</i>	<i>Size</i>	<i>σ/σ_1 range</i>
1	0–51	52	1 to 10^{-1}
2	52–95	44	10^{-3} to 10^{-4}
3	96–111	16	10^{-7} to 10^{-8}
4	112–115	4	10^{-11} to 10^{-13}
Noise	116–155	40	$< 10^{-15}$

The inter-tier gaps span 10^2 – 10^4 in the SV ratio, making the decomposition unambiguous.

Table 1 shows the tier boundaries at representative configurations for both potentials. The first three boundaries (positions 52, 96, 112) are remarkably stable: they agree between $1/r$ and $1/r^2$ at every tested configuration.

Configuration	Potential	Tier positions	Subspace sizes
Lagrange ($\mu=1, \phi=60^\circ$)	$1/r$	52, 96, 110, 115	52, 44, 14, 5
	$1/r^2$	52, 94, 110, 115	52, 42, 16, 5
Isosceles ($\mu=1, \phi=90^\circ$)	$1/r$	52, 96, 112, 115	52, 44, 16, 3
	$1/r^2$	52, 96, 112, 114	52, 44, 16, 2
Large- μ Lagrange	$1/r$	52, 96, 112, 116	52, 44, 16, 4
	$1/r^2$	52, 96, 112, 116	52, 44, 16, 4
Collinear ($\phi \approx 0^\circ$)	$1/r$	151	151
	$1/r^2$	151	151

Table 1: Tier boundaries at special configurations ($\varepsilon = 10^{-4}$). The “ideal” decomposition $52 + 44 + 16 + 4$ emerges cleanly at large- μ Lagrange; minor variations (± 2 generators) occur at exact symmetry points. The collinear configuration is anomalous: all 151 candidate generators become significant.

3.2 Statistical significance

Across the full 100×100 atlas (10,000 grid points, both potentials, 48,318 total tier detections), the preferred tier positions cluster sharply at indices 52, 96, 116, and 129. The Kolmogorov–Smirnov test against a uniform distribution gives $D = 0.667$, $p < 10^{-4}$.

The subspace-size distribution has concentration index (one minus normalised entropy) $c = 0.254$. In 10,000 Monte Carlo trials with uniformly random sizes in $[1, 156]$, the maximum concentration achieved was $c \approx 0.18$. The observed quantization of tier sizes is statistically extreme ($p < 10^{-4}$).

The most frequent subspace sizes (Table 2) coincide with the tier sizes and their complements to 116.

Size	Count	Fraction
116	13,758	28.5%
4	5,538	11.5%
112	2,625	5.4%
52	1,202	2.5%
44	—	—
16	1,374	2.8%

Table 2: Most frequent subspace sizes across 48,318 tier detections. The size 116 dominates because at many configurations only the main cliff is detected.

4 S_3 representation theory

The symmetric group S_3 acts on the three-body configuration space by permuting the labels $(1, 2, 3)$. For equal masses this is a genuine symmetry of the Hamiltonian; for unequal masses it is a symmetry of the algebraic structure (which monomials appear in the generators), even though the numerical coefficients differ [1].

4.1 Character table and Clebsch–Gordan rules

S_3 has three irreducible representations:

Irrep	dim	Character on (12)	Character on (123)
A (trivial)	1	1	1
A' (sign)	1	−1	1
E (standard)	2	0	−1

The Clebsch–Gordan decomposition rules are:

$$\begin{aligned}
A \otimes A &= A, & A \otimes A' &= A', & A \otimes E &= E, \\
A' \otimes A' &= A, & A' \otimes E &= E, & E \otimes E &= A \oplus A' \oplus E.
\end{aligned}$$

The exterior square relevant for self-brackets is $\Lambda^2(E) = A'$.

4.2 Isotypic decomposition of the algebra

The three pairwise Hamiltonians $\{H_{12}, H_{13}, H_{23}\}$ form the edge representation of S_3 , which decomposes as $A \oplus E$. At level 1, the three tidal operators $K_i = \{H_{jk}, H_{j\ell}\}$ (one per vertex) also form $A \oplus E$. Higher levels are generated by the Clebsch–Gordan rules.

Theorem 2 (Isotypic decomposition). *The 156 candidate generators of $\mathcal{A}$ through bracket level 3 decompose under S_3 as*

$$24 A \oplus 28 A' \oplus 104 E \quad (\text{total dimension } 24 + 28 + 104 = 156).$$

At each bracket level, the decomposition is:

Level	n_A	$n_{A'}$	n_E	dim	E -fraction
0	1	0	1	3	66.7%
1	1	0	1	3	66.7%
2	2	2	4	12	66.7%
3	20	26	46	138	66.7%
Total	24	28	104	156	66.7%

The E -fraction is exactly $2/3$ at every bracket level.

Proof. The seed $\mathcal{A}_0 = A \oplus E$ has E -fraction $2/3$. The key observation is that the Clebsch–Gordan decomposition of an S_3 -module with E -fraction $2/3$ preserves this fraction. At each level n , new generators arise from two sources:

1. *Cross-brackets* $\{\mathcal{A}_{n-1}, \mathcal{A}_k\}$ for $k < n-1$: the tensor product of two modules with E -fraction $2/3$ again has E -fraction $2/3$ (verified by direct computation using the CG rules).
2. *Self-brackets* $\Lambda^2(\mathcal{A}_{n-1})$: the exterior square $\Lambda^2(nA \oplus mA' \oplus \ell E)$ has E -fraction $2/3$ when the input has E -fraction $2/3$ and $n+m=\ell$ (which holds at every level by induction, since $n_A + n_{A'} = n_E$ for the seed and the CG rules preserve this).

We verified the decomposition computationally:

- Level 2: $\{\mathcal{A}_1, \mathcal{A}_0\} = (A+E) \otimes (A+E) = 2A + A' + 3E$ (dim 9), $\Lambda^2(\mathcal{A}_1) = \Lambda^2(A+E) = A' + E$ (dim 3), total $2A + 2A' + 4E = 12$. **Matches observed count.**
- Level 3: $\{\mathcal{A}_2, \mathcal{A}_0\} = 6A + 6A' + 12E$ (dim 36), $\{\mathcal{A}_2, \mathcal{A}_1\} = 6A + 6A' + 12E$ (dim 36), $\Lambda^2(\mathcal{A}_2) = 8A + 14A' + 22E$ (dim 66), total $20A + 26A' + 46E = 138$. **Matches observed count.** $\square$

4.3 The number 52

Corollary 3. *The first tier size equals the number of E -type irreducible copies:*

$$|\text{Tier 1}| = n_E = 52.$$

This identity is exact, not approximate. The 52 E -doublets contribute $2 \times 52 = 104$ generators to the algebra. At a generic (non-symmetric) configuration, the S_3 symmetry is broken: each E -doublet splits into two singlets with distinct singular values. The more dynamically significant component of each doublet enters Tier 1, producing exactly 52 entries.

Remark 4 (Doublet signatures). At the near-Lagrange configuration ($\mu = 0.992$, $\phi = 60.2^\circ$), the S_3 symmetry is approximately restored. We detect near-degenerate doublets (SV ratio < 1.05) in the spectrum:

Tier	Doublets	Singlets	Total
1	18	16	52
2	11	22	44
3	2	12	16
4	0	4	4

The high doublet count in Tier 1 confirms the E -doublet origin. Full degeneracy would require exact S_3 symmetry; the splitting reflects the grid’s approximation to the Lagrange point.

4.4 Tier structure as broken isotypic decomposition

At the S_3 -symmetric Lagrange point, the E -type generators carry the most dynamical information because they sense the *relative geometry* of the three bodies—they transform nontrivially under permutations. The A -type generators measure *total* (symmetric) quantities, and A' -type generators measure *antisymmetric* combinations.

The tier structure reflects this hierarchy:

- **Tier 1 (52):** One component from each of the 52 E -doublets—the dynamically dominant direction.
- **Tier 2 (44):** The partner component from 44 of the 52 doublets. The 8 missing partners have fallen below the Tier 2 boundary (severe symmetry breaking displaces one partner).
- **Tiers 3 and 4 (16 + 4 = 20):** A subset of the $n_A + n_{A'} = 24 + 28 = 52$ generators in one-dimensional representations.

- **Noise (40):** The remaining $52 - 20 = 32$ generators from A/A' irreps and 8 displaced E -partners, collectively forming the syzygy space (Section 6).

Remark 5. The decomposition $n_A + n_{A'} = 52 = n_E$ is not a coincidence: it follows from the CG rules applied to the seed $A \oplus E$ and the fact that the E -fraction is locked at $2/3$ at every level. Since each E contributes dimension 2, the A/A' count $n_A + n_{A'}$ equals the E count n_E .

5 Jet filtration and integer scaling

The tier decomposition has a natural interpretation in terms of jet spaces. Define the k -jet of a smooth function f at a configuration q as its Taylor polynomial through order k :

$$j_q^k f = \sum_{|\alpha| \leq k} \frac{1}{\alpha!} \left(\frac{\partial^{|\alpha|} f}{\partial z^\alpha} \right) \Big|_q (z - q)^\alpha,$$

where z denotes phase-space coordinates. The evaluation matrix M constructed by sampling in an ε -ball captures the Taylor expansion of each generator up to terms of order $O(\varepsilon^K)$ for some effective truncation order K . The singular values of M should then scale as $\sigma_k \sim \varepsilon^{\alpha_k}$ where α_k is the lowest Taylor order at which the k -th independent direction contributes new information.

5.1 Scaling exponents

Theorem 6 (Integer scaling). *At the Lagrange configuration, fitting $\log_{10} \sigma_k$ against $\log_{10} \varepsilon$ across five sampling radii gives scaling exponents $\alpha = \alpha_k$ that are integer-quantized:*

Component	Count	$\bar{\alpha} \pm \delta\alpha$	Nearest integer	Interpretation
Tier 1	52	-0.007 ± 0.014	0	zeroth-order observables
Tier 2	44	1.002 ± 0.025	1	first-order variation
Tier 3	16	2.005 ± 0.165	2	second-order variation
Tier 4	4	2.824 ± 0.210	3	third-order variation
Noise	40	flat at $\sim 10^{-15}$	∞	identically zero

The noise floor shows no ε -dependence because the corresponding linear combinations are exactly zero.

Proof (numerical). For each SV index k and each of the five sampling radii ε_j , the singular value $\sigma_k(\varepsilon_j)$ is recorded at the Lagrange configuration ($\mu = 1, \phi = 60^\circ$). Least-squares regression of $\log_{10} \sigma_k$ against $\log_{10} \varepsilon_j$ yields the slope α_k and coefficient of determination R^2 . For indices $0 \leq k \leq 115$ (the 116 significant directions), $R^2 > 0.99$ in all cases, confirming the power-law relationship. The mean exponents by tier are as stated above.

For the noise floor ($k \geq 116$), the singular values are at machine epsilon ($\sim 10^{-16}$) regardless of ε , confirming that these directions are identically zero. $\square$

Remark 7 (Jet filtration). Define the k -th jet subspace $\mathcal{J}_q^k \subset \mathcal{A}_3$ as the span of generators whose local Taylor expansion around q begins at order $\leq k$. Theorem 6 shows that

$$\mathcal{J}_q^0 \subset \mathcal{J}_q^1 \subset \mathcal{J}_q^2 \subset \mathcal{J}_q^3 = \mathcal{A}_3|_q$$

with

$$\dim \mathcal{J}_q^0 = 52, \quad \dim \mathcal{J}_q^1 = 96, \quad \dim \mathcal{J}_q^2 = 112, \quad \dim \mathcal{J}_q^3 = 116.$$

The successive quotients have dimensions 52, 44, 16, 4, matching the tier sizes. This jet filtration refines the bracket-level filtration $\mathcal{A}_0 \subset \mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{A}_3$ (which has dimensions 3, 6, 17, 116) and is orthogonal to it: generators from the same bracket level can appear in different jets.

5.2 Gap growth

The gap ratio at the rank boundary (SV index 116) grows with ε :

ε	$\sigma_{115}/\sigma_{116}$
10^{-4}	5.5
2×10^{-4}	51
5×10^{-4}	950
10^{-3}	8,100
2×10^{-3}	67,472

The growth is approximately $\propto \varepsilon^2$, consistent with Tier 4 generators scaling as ε^3 while the noise floor remains constant. At $\varepsilon \geq 10^{-3}$ the rank determination is definitive (gap $> 10^3$).

5.3 Relationship between jet and bracket filtrations

The bracket-level filtration and the jet filtration are distinct. The bracket filtration has dimensions 3, 6, 17, 116 and is purely algebraic (determined by bracket depth). The jet filtration has dimensions 52, 96, 112, 116 and depends on the *local* Taylor expansion around a configuration.

Levels 0 and 1 (6 generators total) contribute only to Tier 1, as expected: the Hamiltonians and their first brackets are zeroth-order observables. Level 2 generators (11 new) appear in Tiers 1 and 2. Level 3 generators (99 new) populate all four tiers. The jet filtration thus provides finer information than the bracket depth.

6 Syzygy structure of the null space

The 40 directions in the SV spectrum below the rank-116 cliff require careful interpretation. If they corresponded to generators that vanish identically on phase space, they would be Dirac-type constraints and the tier structure might reflect a first-class/second-class distinction rather than an isotypic decomposition.

6.1 No generator vanishes individually

Proposition 8. *All 156 candidate generators of $\mathcal{A}_3$ are individually nonzero as functions on $\mathbb{R}^{12}$. At every tested configuration (Lagrange, isosceles, collinear, scalene) and every sampling radius $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}\}$, the RMS value of each generator exceeds 10^{-14} .*

Proof (computational). The symbolic Poisson algebra was built through level 3, producing 156 generators as symbolic expressions. Each was evaluated independently at 500 random phase-space points in an ε -ball around each of four special configurations. The column RMS $(\sum_i g(z_i)^2/N)^{1/2}$ exceeded 10^{-14} for all generators at all configurations and all ε . $\square$

The 40 null directions in the SVD are therefore *syzygies*: linear combinations $\sum_k c_k g_k = 0$ of generators that cancel, not individual generators that vanish.

6.2 Deep and soft syzygies

The 40 syzygies decompose into two populations:

Definition 9. *A syzygy is deep if the corresponding singular value is at machine epsilon ($< 10^{-14}$) at every configuration and every ε . A syzygy is soft if it holds at generic configurations but breaks at special submanifolds.*

Proposition 10. *The 40-dimensional null space of $\mathcal{A}_3$ decomposes as*

$$\underbrace{8 \text{ deep syzygies}}_{\text{universal identities}} \oplus \underbrace{32 \text{ soft syzygies}}_{\text{Jacobi consequences}}.$$

1. *The 8 deep syzygies (SV indices 148–155) satisfy $\sigma < 2.3 \times 10^{-14}$ across all 10,000 configurations and all 5 sampling radii. Indices 151–155 are at machine epsilon ($< 10^{-15}$) everywhere.*
2. *The 32 soft syzygies (SV indices 116–147) hold at generic configurations but break near the collinear submanifold ($\phi \approx 6^\circ$): 8 of these “wake up,” increasing the effective rank from 116 to 124. The atlas-wide maximum reaches 125 (9 awake) at a single grid cell, ($\mu = 1.020, \phi = 5.7^\circ$).*

Remark 11. The soft syzygies are consequences of the Jacobi identity $\{A, \{B, C\}\} + \text{cyc} = 0$. At generic configurations, the three cyclic terms cancel. At collinear configurations, where certain brackets degenerate, the cancellation fails and formerly dependent directions become independent.

6.3 Null-space structure

Analysis of the right singular vectors v_k for $k \geq 116$ reveals:

1. **Level concentration.** 96.4% of the null-space weight is in level-3 generators, $\sim 3.4\%$ in level 2, and 0% in levels 0–1. No seed or first-bracket generator participates in any syzygy.
2. **Cross-level mixing.** 30–39 of the 40 null vectors mix generators from different bracket levels. These are *cross-level syzygies*: algebraic identities relating generators at different bracket depths.
3. **Potential independence.** Both $1/r$ and $1/r^2$ give exactly 40 null directions at every configuration. The syzygies are a property of the bracket structure and the three-body interaction graph, not the potential.

6.4 Orthogonality of tiers and syzygies

The tier decomposition and the syzygy structure are orthogonal:

$$\underbrace{156 \text{ generators}}_{\text{all nonzero}} = \underbrace{116 \text{ independent}}_{\text{organised into 4 tiers by } S_3} \oplus \underbrace{40 \text{ syzygies}}_{\text{algebraic identities}}.$$

The tier boundaries reflect *dynamical significance* mediated by S_3 representation theory: which generators carry the most geometric information at a given configuration. The syzygy boundary reflects *algebraic redundancy*: which generators are expressible as linear combinations of others. These are independent classifications.

Proposition 12. *The modal rank is 116 at all thresholds $\geq 10^{-10}$. The rank is stable:*

Threshold	Min rank	Modal rank	Max rank
10^{-8}	108	112	116
10^{-10}	114	116	125
10^{-12}	116	116	138
10^{-13}	116	116	148

Configurations with rank > 116 cluster near the collinear submanifold where soft syzygies break.

7 Discussion

7.1 Intrinsic quantization

The tier sizes 52, 44, 16, 4 do not correspond to any known quantization scheme. We tested explicitly:

- **Loop quantum gravity** (area spectrum $\sqrt{j(j+1)}$ for half-integer j): mean matching error 0.153, Monte Carlo $p = 0.77$ (not significant).
- **Bekenstein** (equally-spaced integer areas): mean matching error 0.093, $p = 0.18$ (not significant).
- **SU(2) representation dimensions** ($2j+1$): no correspondence.

The quantization is intrinsic: it arises from the S_3 isotypic decomposition of the Poisson algebra, not from any external quantum-gravitational structure.

7.2 Physical interpretation of E-dominance

The E -type generators are dynamically dominant because they sense the *relative geometry* of the three bodies. In the standard representation E , S_3 acts by permuting the vertices of an equilateral triangle; the E -components of a function measure how the function value changes when the particle labels are permuted. At a generic (scalene) configuration where all three bodies are distinguishable, these are the most informative observables.

The A -type generators measure *total* quantities invariant under permutation, and A' -type generators measure fully antisymmetric combinations. Both are less informative at generic configurations because they do not distinguish which body is which.

7.3 Open questions

1. **Level 4 tier structure.** Does the jet filtration extend to level 4? The lower bound $d(4) \geq 5,604$ (200,000 samples, not definitive) from [1] implies a vastly larger algebra. Whether the four-tier pattern persists, or whether new jet orders appear, is unknown.
2. **$N = 4$ analogue.** The four-body dimension sequence [6, 14, 62] (through level 2) has been computed [1]. The relevant symmetry group is S_4 , with five irreps. Whether an analogous tier decomposition exists—and whether its sizes are predicted by S_4 Clebsch–Gordan rules—is a natural question.
3. **Jet-order growth.** The exponents $\alpha = 0, 1, 2, 3$ match the bracket levels 0, 1, 2, 3 in number but not in dimension (the bracket filtration has dimensions 3, 6, 17, 116 while the jet filtration has dimensions 52, 96, 112, 116). Is there a representation-theoretic formula predicting $\dim \mathcal{J}_q^k$ in terms of the S_3 content at each bracket level?
4. **Syzygy resolution.** The 8 deep syzygies likely correspond to consequences of translational symmetry (total momentum conservation). Can they be identified explicitly as expressions proportional to P_x and P_y brackets at various levels?
5. **Collinear anomaly.** Why do 8 soft syzygies break specifically at near-collinear configurations? The collinear arrangement is an S_2 -symmetric stratum (one pair of bodies defines a distinguished axis); the syzygy-breaking may reflect the reduction $S_3 \rightarrow S_2$ and the branching rules for the sign representation.

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References

- [1] Brian J. Sheppard, *Super-exponential growth of the Poisson algebra generated by pairwise Hamiltonians of the planar three-body problem*, Preprint, 2026.
- [2] Anthropic, *Claude (Opus 4.6)* [Large language model], 2026. <https://claude.ai/>
- [3] J.-P. Serre, *Linear Representations of Finite Groups*, Springer, 1977.
- [4] W. Fulton and J. Harris, *Representation Theory: A First Course*, Springer, 1991.
- [5] J. Kollár, *Lectures on Resolution of Singularities*, Ann. of Math. Stud., vol. 166, Princeton Univ. Press, 2007.
- [6] J. Moser, *Three integrable Hamiltonian systems connected with isospectral deformations*, Adv. Math. **16** (1975), 197–220.
- [7] F. Calogero, *Solution of the one-dimensional N -body problems with quadratic and/or inversely quadratic pair potentials*, J. Math. Phys. **12** (1971), 419–436.
- [8] P. A. M. Dirac, *Generalized Hamiltonian dynamics*, Canad. J. Math. **2** (1950), 129–148.
- [9] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton Univ. Press, 1992.